

MCB

Inventory Control System

PD Meeting Summary • Workflow • Controls • Decisions

Purpose	PD review and workflow design freeze
Baseline	MCB Inventory System FINAL v94
Working candidate	v95 workflow-lock candidate
Review date	28 August 2026

Prepared as a management-ready summary plus implementation control specification. Unresolved business authority is explicitly marked NEEDS DECISION rather than guessed.

1. Executive summary

MCB Inventory Control is designed as one centralized, multi-site construction inventory system. Its purpose is to make material movement controlled, traceable and accountable - not simply to replace Excel with a web form.

1 shared system	Strict BOM	Controlled stock	Audit first
Multi-site, multi-user source of truth	No operational over-BOM override	Current Stock changes only through valid movement	Posted history is preserved

What management gets

- Visibility of stock, low/out conditions, BOM exposure, equipment status and exceptions by site.
- Clear responsibility for receiving, issuing, transferring, counting, correcting and closing stock.
- System blocking of insufficient stock and over-BOM transactions before they damage inventory accuracy.
- Immutable posted history with correction/reversal rather than delete-and-hide behavior.
- A phased path from manual site forms to AI/Telegram-assisted data capture without allowing AI to post stock directly.

Operating principle

Normal work should be fast and system-validated. Human control is concentrated at high-risk points and exceptions. This avoids turning the Stock Controller into an approval bottleneck for every routine line.

2. Core process map

Process	Normal owner	System control	Stock timing
Item Master	Developer/Admin per permission	Global identity; Active/Inactive; no automatic zero rows at all sites	None
Stock In	Stockkeeper -> Stock Controller	DN/evidence + Approved BOM	Only after verification/post
Stock Out	Stock Controller daily batch	Available Stock + Approved BOM; aggregate batch validation	On batch Post
Site Transfer	Sender -> Receiver	Reservation, dispatch, receipt, mismatch	Reserve -> dispatch -> receipt
Equipment	Site custody users	Checkout/outstanding/return/condition	Custody state
Physical Count	Stockkeeper + Stock Controller (+ Admin monthly)	Blind count, recount variance, controlled adjustment	Only after variance approval
Correction	Authorized controller	Original is immutable; linked correction/reversal	New audited movement

3. Responsibility model

Role	Primary responsibility	Default limits
Developer	System administration; role/site/permission control; technical recovery	Cannot erase audit history; critical override should require reason
Stock Controller	Daily inventory control; Stock Out; Stock In verification; counts; serious exceptions	No BOM override; no direct Current Stock typing; no history deletion
Stockkeeper	Receive/check/submit Stock In; transfer/receipt; equipment; count entry	No Stock Out posting by default; no adjustment/BOM override
QS	Approved BOM/cost-code control; BOM revision preparation	No stock posting
Admin	Site/master administrative support; monthly joint count	No self-escalation of user access
Management	Dashboard/report/audit oversight	Read-oriented by default

4. Decisions required from PD / management

LOCKED	NEEDS DECISION	BUILD GAP	FUTURE PHASE
Topic	Decision needed	Status	
BOM revision	Who gives final approval after QS prepares an increase?	NEEDS DECISION	
Transfer BOM timing	When allocation moves: approval, dispatch or receipt?	NEEDS DECISION	
Count variance	Who authorizes final inventory adjustment?	NEEDS DECISION	
Month close	Who can close/reopen the inventory period?	NEEDS DECISION	
Transfer approval	Who approves normal site-to-site transfer?	NEEDS DECISION	
PO/DN exceptions	Allow no-PO receipt? Rules for substitution/over-delivery/cancelled PO?	NEEDS DECISION	
Equipment exceptions	Who concludes Lost/Damaged/Repair/Scrap/replacement?	NEEDS DECISION	

5. Technical readiness

The protected source baseline is FINAL v94. A v95 workflow-lock candidate is being hardened against the agreed business specification. It should be treated as a PD review / development candidate until production gates are completed.

Production gate before broad launch

- Relational/ledger posting for stock, BOM, reservation and transfer rather than whole-system JSON as the long-term production core.
- Atomic posting and idempotency so retries or simultaneous users cannot double-post or consume the same balance.
- Full transfer lifecycle including Approved/Reserved, Dispatch, partial receipt and disputed in-transit quantity.
- Permission Overrides enforced server-side, not only hidden UI actions.
- Blind-count/recount flow, count variance authorization, month closing and site lifecycle enforcement.
- Backup/restore procedure tested against runtime data, not only source-code ZIP backup.

6. Rollout recommendation

Phase	Focus	Exit condition
1 - Controlled manual pilot	Real site users, real forms, strict BOM/stock rules	Daily flow accurate and users trained
2 - Stabilize	Counts, transfer exceptions, reports, period control	Acceptance tests pass on pilot sites
3 - AI assistance	Telegram/photo extraction into review grid	Human verification remains mandatory
4 - Multi-site expansion	Scale sites/users after concurrency/security tests	Production architecture and recovery tests pass